development/testing and stabilising of applications could however be done in the cloud. This approach greatly reduces the cost and time associated with providing infrastructure for development and testing, and helps enterprises in not having to be saddled with a perpetual investment in servers, machines and software.

Hence, in the context of the public cloud, you definitely have a choice of deployment models and it is not an 'All or nothing' scenario.

13. Embrace native services: Don't be a fence sitter

The cloud platforms today are seeing a rapid pace of continuous innovation where, increasingly, the power behind these innovations comes not from scratching the surface of infrastructure services, but from deep down domain-specific solution areas. While these capabilities had not matured till the recent past, and enterprises were content with evaluating the cloud for just what it offered at the surface, they ended up deploying their applications, as is, on infrastructure services like compute and networking. This approach also enabled them to avoid getting locked-in to a particular vendor, and obviated the need to make changes to their applications when deploying to a different cloud host. However, enterprises are fast realising that this approach, while good for a first step, was not enough to catapult them into a league where their businesses could really benefit from the full impact of the cloud. This next level of engagement with the cloud platform can be achieved by making the following considerations:

- Architect and build your applications to utilise the native capabilities in the cloud platform, which are tried and tested for scale, performance and reliability. Expunge custom code and components within the application, where there are equivalent solutions that are much more conformant to industry standards and which have been battle hardened.
- Leading cloud platforms support a wide range of platforms and technologies, be it open source or otherwise. Hence, enterprises can continue to operate in a familiar environment using tools they are already comfortable with. At the same time, they can seamlessly embed the native features to reap the benefits of embracing the cloud.
- Tried and tested native capabilities in the cloud ought to be considered to replace custom or vendor-specific implementations of known problems. By doing so, you can dedicate time to innovate on the solution domain instead of the solution host/infrastructure.

Though native services are proprietary offerings by a cloud vendor, they are mostly built using proven techniques and platforms. As long as they are built using open standards and provide you a choice and familiarity, you should definitely consider them.

14. You need to re-benchmark your infrastructure

Capacity planning is an activity one undertakes to design the host environment in terms of RAM, CPU, cores and eventually, the number and type of machines required. Test and staging environments are a reflection of the production environment and, hence, you can arrive at a benchmark or statistics with respect to performance. Solutions are tweaked and optimised for specific kinds of hardware.

In this context, when you plan moving to the cloud, at the outset, you will not get the exact replica of the configurations. Hence, it is highly recommended to assess performance or re-benchmark against the new hardware specifications available in the cloud.

15. Assess for security and compliance

Dynamics change in the context of the public cloud, more so when you are choosing a region that falls outside a specific geographic boundary. Irrespective of whether your solution comes under the purview of data protection laws and privacy regulations, or adherence to global standards on the compliance front, it is important to assess the host environment's coverage on the following fronts:

- Security
- Privacy
- Transparency
- Compliance

In order to leverage the benefits of hosting in the cloud, you have to believe in the cloud providers' claims about adhering to laws or regulations pertaining to your domain and entrust them with managing your data. As an example, on the compliance front, it is absolutely essential to ensure the cloud provider conforms to industry defined global standards (like FISMA, HIPPA, FERPA, etc) as well as country-specific standards (like NZ GCIO, UK G-Cloud, etc).

Summary

At the end of the day, moving into the cloud gives you an opportunity to operate with unlimited compute and, hence, no boundary. It is absolutely critical to 'do it right the first time' and optimise on the efforts. At times, taking up migration to the cloud has resulted in rejigging in teams and their operating model. Awareness about the cloud platform, complemented by the above guidelines, will help you arrive at the starting line in terms of the overall planning cycle of your transformation journey. While the cloud will give you a sophisticated compute environment, planning is key to success. Before you move to the cloud, understand its potential. END

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